

To the Editors, *Advances in Data Analysis and Classification*

Dear Editors,

I submit the manuscript “A crossed random-effects latent class model for annotation ensembles of large language models” for consideration in *Advances in Data Analysis and Classification*.

The paper addresses a problem that has appeared quickly and widely: researchers now label text with several large language models, several prompts and repeated runs, and aggregate the labels with Dawid–Skene-type models that assume the raters err independently. Configurations built on shared corpora and alignment do not. I propose a latent class model whose conditional-dependence structure is derived from the factorial design of the annotation ensemble (model $\times$ prompt $\times$ run), estimated by Bayesian methods with the class marginalised. The model nests Dawid–Skene and the random-effects latent class model of Qu, Tan and Kutner; a simulation study shows that independence-based aggregation overstates accuracy and biases prevalence monotonically in the strength of dependence, that the proposed model recovers the parameters without a gold standard, and that its variance decomposition gives direct guidance on how many models, prompts and runs to use. The paper characterises a merged-class posterior mode that defeats naive estimation and shows two remedies. An application to policy-topic coding of US congressional bill titles, with human codes from the Comparative Agendas Project held out for validation, illustrates the method.

The work sits in the journal’s tradition of relaxing local independence in latent class and mixture models (Oberski, van Kollenburg and Vermunt 2013; Oberski 2016; Marbac, Biernacki and Vandewalle 2016) and of modelling annotator heterogeneity (Salter-Townshend and Murphy 2014). All code, prompts, model version identifiers and annotations are archived with a DOI. The manuscript is not under consideration elsewhere, and I have no competing interests to declare.

Suggested reviewers: [to be added — three names from different countries with institutional e-mails; candidates from the latent class / rater agreement literature and from computational social science].

Yours sincerely,

[name, affiliation, ORCID]